

LESSON 7-6 • SESSION 1 • PRACTICE • FORM A

Distance Between Points on a Grid

Learning target – I can find the distance between two points that share a coordinate. · 6.NS.C.8

REMEMBER

- Same **y-coordinate**: the points are side to side. Use the x-coordinates.
- Same **x-coordinate**: the points are up and down. Use the y-coordinates.
- **Opposite signs**: add the absolute values. **Same sign**: subtract them. $(5, -2)$ to $(5, -9)$ is $9 - 2 = 7$.

WORKED EXAMPLE

Each unit is 1 meter. Find the distance from $P(-9, 4)$ to $Q(3, 4)$, and from P to $R(-9, 11)$.

Step 1 P and Q share $y = 4$, so use x . -9 and 3 have opposite signs: $9 + 3 = 12$ m.

Step 2 P and R share $x = -9$, so use y . 4 and 11 are both positive: $11 - 4 = 7$ m.

Answer $PQ = 12$ m and $PR = 7$ m

WORDS distance (*distancia*) · absolute value (*valor absoluto*) · coordinate plane (*plano de coordenadas*)

PRACTICE

show your work · label units

1. Find the distance between each pair of points.

a. $(3, 14)$ and $(3, 4)$ _____

b. $(-11, 2)$ and $(6, 2)$ _____

c. $(-13, 9)$ and $(-13, -6)$ _____

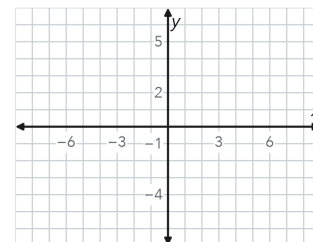
d. $(-6, 4)$ and $(5, 4)$ _____

2. **Town Map Walk.** Each unit is 1 kilometer. Library $(-8, 5)$, Pool $(7, 5)$, Clinic $(-8, -6)$.

a. Plot and label the three places.

b. Library to Pool _____ km

c. Library to Clinic _____ km



3. **Drone Flight.** A drone flies straight from $(-5, 13)$ to $(-5, -4)$. Each unit is 1 meter. How far does it fly?

Hint Same x , so use y . 13 and -4 have opposite signs.

Show your work.

Answer _____ m

4. **One Quadrant.** Find the distance from $(-16, -7)$ to $(-16, -20)$.

Hint Both y -values are negative, so subtract.

Show your work.

Answer _____ units

5. **Missing Point.** Point G is at $(10, -3)$. Point H is 15 units directly to the left of G . Name point H .

Hint Moving left changes only x : $10 - 15$.

Show your work.

H = _____

6. **Window Washer.** A window washer moves from $(21, 18)$ to $(21, 7)$. Each unit is 1 foot. How far does the washer move?

Hint Are 18 and 7 on the same side of 0 ?

Show your work.

Answer _____ ft

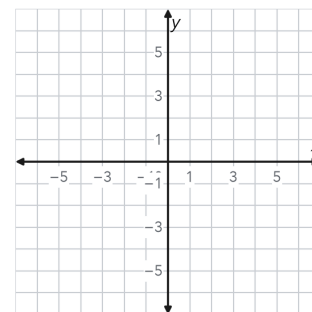
7. **Bus Route.** A bus runs straight from $(-17, -4)$ to $(12, -4)$. Each unit is 1 block. How far does the bus go?

Show your work.

Answer _____ blocks

8. **City Grid.** Each unit is 1 block. A library is at $(-4, 6)$, a school at $(5, 6)$ and a park at $(5, -3)$. Plot the walk on the grid.

- a. Library to school _____ blocks
b. School to park _____ blocks
c. Total walk _____ blocks



FIND THE ERROR

explain the mistake, then fix it

9. **Malik's Work.** To find the distance from $(2, -9)$ to $(2, 6)$, Malik wrote $9 - 6 = 3$ units. Explain his mistake. Then find the correct distance.

TEST PRACTICE

choose the best answer

10. The distance from $(4, 2)$ to $(4, 9)$ is –
A 2 units **B** 7 units **C** 11 units **D** 13 units
11. The distance from $(-3, 5)$ to $(6, 5)$ is –
A 3 units **B** 9 units **C** 11 units **D** -9 units

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

(-5, -3) N	17 L	27 Q	25 J	18 H	13 E	11 G	5 X	29 T	(25, -3) Z
		#3	#4	#5	#6	#7	#8c		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____